

Golden Rule -> similar meaning -> similar vectors -> nearby points in vector space

1. What Are Embeddings? (RAG Recap)

An embedding is a vector representation of text such that semantically similar text ends up with vectors that sit close together in space. In RAG, both the document chunks and the user's query get embedded, so the system can compare the query vector against every chunk vector and return the closest matches.

2. Popular Embedding Models

Model	Provider	Notes
text-embedding-3-small / large	OpenAI	Easy API access, strong general-purpose performance
all-MiniLM-L6-v2	sentence-transformers (open-source)	Free, runs locally, smaller vector dimension (384)
embed-v3	Cohere	Strong multilingual support

3. Popular Vector Stores

Store	Type	Notes
FAISS	Local, in-memory/on-disk library	Free, fast, no server needed; great for prototypes
Chroma	Local / embedded database	Easy persistence; good fit for small-to-medium projects
Pinecone	Managed cloud service	Scales to production; infrastructure is handled for you

Also common in production stacks: Weaviate and Milvus.

4. Similarity Metrics

- **Cosine similarity** — measures the angle between two vectors; the most common choice for text embeddings.
- **Dot product** — similar to cosine but sensitive to vector magnitude.
- **Euclidean distance** — straight-line distance; less common for normalized text embeddings.

5. Code Example

```
$ from langchain_openai import OpenAIEmbeddings
$ from langchain_community.vectorstores import FAISS

$ embeddings = OpenAIEmbeddings()
$ vectorstore = FAISS.from_documents(chunks, embeddings)
$ vectorstore.save_local("faiss_index")
```

Pitfall

Mixing embeddings from two different models inside the same vector store silently breaks similarity search — the vectors aren't in the same space, so "closeness" stops meaning anything.

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Remember

Store metadata (source, page, chunk id) alongside every vector so retrieved chunks can always be traced back and cited.

INTERVIEW PREP

Q: What does it mean for two embeddings to be “close”?

A: Their vectors have a small angle or distance (commonly measured with cosine similarity) in the embedding space, indicating the underlying text is semantically similar.

Q: Why can't you swap embedding models on an existing vector store without re-embedding?

A: Different models map text into different vector spaces with different dimensions and geometry, so a query embedded with model B won't compare meaningfully against document vectors embedded with model A.

Q: FAISS vs Pinecone — how do you choose?

A: FAISS is a free, local library well suited to prototyping or small-to-medium local apps; Pinecone is a managed cloud vector database better suited to production scale, high availability, and teams that don't want to manage infrastructure.

Q: What similarity metric is most commonly used for text embeddings, and why?

A: Cosine similarity, because it measures the angle between vectors regardless of magnitude, which suits normalized text embeddings well.

Q: What gets stored in a vector store besides the embedding vector itself?

A: The original chunk text (or a reference to it) plus metadata such as source document, page number, and chunk id, so retrieved results can be used and cited.